

DTSA 5735

Advanced Topics and Future Trends in Database Technologies

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (40.7 / 100)	32.0 / 100
SPECIALIZATION Database Architecture, Scale, and NoSQL		

32

CURRENT AVG SCORE

across 6 job roles

84

POTENTIAL SCORE

with recommended additions

+63

MAX SCORE GAIN

best single-role improvement

CURRENT COURSE DESCRIPTION

The course will have an overview of future trends in databases, including non-relational databases (NoSQL) and Big Data. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	40.7	86.3	+45.6	HIGH
ML Engineer	38.7	87.9	+49.2	HIGH
Data Scientist	33.0	89.1	+56.2	HIGH
MLOps Engineer	30.7	93.9	+63.2	HIGH
Machine Learning Engineer	27.0	74.6	+47.6	HIGH
Applied Scientist	21.7	72.6	+50.9	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

LLMs & Generative AI 2 of 6 roles

storage

Model Deployment & Serving 2 of 6 roles

production

Business Communication 1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

3 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

4 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

AI Engineer

40.7 → 86.3 (+45.6 pts)

engineering models learning build form software systems storage

ML Engineer

38.7 → 87.9 (+49.2 pts)

learning systems models production engineering starting world build

Data Scientist

33.0 → 89.1 (+56.2 pts)

product characteristics analytics evolution insights different build engineering

MLOps Engineer

30.7 → 93.9 (+63.2 pts)

production systems learning engineering tools models

Machine Learning Engineer

27.0 → 74.6 (+47.6 pts)

engineering learning analytics tools solutions time design science

Applied Scientist

21.7 → 72.6 (+50.9 pts)

lab storage diffusion interdisciplinary models learning systems engineering

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.